Adding and multiplying inequalities

Two inequalities pointing the same way can be added — and, when everything is positive, multiplied.

LESSON 42–43 | 2 × 40 MIN | ALGEBRA 8, §13 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Add two inequalities of the same direction.
- Multiply two inequalities of the same direction when all terms are positive.
- Estimate the range of a sum, difference, product or quotient.
- Explain why inequalities may not be subtracted or divided term by term.

TEXTBOOK

National	§13, pp. 75–79
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Adding

RULE

If $a > b$ and $c > d$ then $a + c > b + d$.

Inequalities pointing the **same way** may be added term by term. The proof is two steps of the last lesson: $a + c > b + c$ (add c) and $b + c > b + d$ (add b), then transitivity.

YOU MAY NOT SUBTRACT THEM

From $5 > 3$ and $10 > 1$ subtracting term by term would give $-5 > 2$, which is false. To handle a difference, reverse the second inequality first and then add: $c > d$ becomes $-c < -d$.

Multiplying

RULE

If $a > b > 0$ and $c > d > 0$ then $ac > bd$.

All four numbers must be **positive**. Drop that condition and the rule collapses: $2 > -5$ and $3 > -4$, but 6 is not greater than 20.

Nor may you divide term by term — invert the second inequality and multiply instead.

Estimating a range

This is what the rules are for. Given $3 < a < 5$ and $2 < b < 4$:

EXPRESSION	WORKING	RANGE
$a + b$	add the two ranges	$5 < a + b < 9$
$a - b$	reverse b: $-4 < -b < -2$, then add	$-1 < a - b < 3$
ab	multiply (all positive)	$6 < ab < 20$
$\frac{a}{b}$	reverse b: $\frac{1}{4} < \frac{1}{b} < \frac{1}{2}$, then multiply	$\frac{3}{4} < \frac{a}{b} < \frac{5}{2}$

The pattern for a difference or a quotient is always the same: **turn the second range round first**, then use the rule you are allowed to use.

02 Worked examples

EXAMPLE 1 ***Given $3 < a < 5$ and $2 < b < 4$, find the range of $a + b$***

1. $3 < a < 5$

   First range.

2. $2 < b < 4$

   Second range, same direction.

3. $3 + 2 < a + b < 5 + 4$

   Add the ends.

4. $5 < a + b < 9$

ANSWER

$5 < a + b < 9$

EXAMPLE 2 ***Given $3 < a < 5$ and $2 < b < 4$, find the range of $a - b$***

1. $2 < b < 4$

   Start from b.

2. $-4 < -b < -2$

   Multiply by -1 — both signs reverse.

3. $3 - 4 < a - b < 5 - 2$

   Now add.

4. $-1 < a - b < 3$

ANSWER

$-1 < a - b < 3$

EXAMPLE 3 ***Given $2 < x < 6$ and $1 < y < 3$, find the range of $\frac{x}{y}$***

1. $1 < y < 3$

2. $\frac{1}{3} < \frac{1}{y} < 1$

   Take reciprocals — the direction reverses.

3. $2 \cdot \frac{1}{3} < \frac{x}{y} < 6 \cdot 1$

   Multiply, all terms positive.

4. $\frac{2}{3} < \frac{x}{y} < 6$

ANSWER

$$\frac{2}{3} < \frac{x}{y} < 6$$

03 Interactive model

Set two boundaries and let the class predict the range of the sum before you show it.

Estimating a range

$3 < a < 5$ and $2 < b < 4$

1. Both point the same way, so they may simply be added.

2. $3 + 2 = 5$

and $5 + 4 = 9$

Add matching ends.

3. $5 < a + b < 9$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Add inequalities	Tengsizliklarni qo‘shish	Сложение неравенств
Multiply inequalities	Tengsizliklarni ko‘paytirish	Умножение неравенств
Same direction	Bir xil yo‘nalish	Одинаковый знак
Estimate	Baholash	Оценить
Range	Oraliq	Промежуток
Lower bound	Quyi chegara	Нижняя граница
Upper bound	Yuqori chegara	Верхняя граница
Term by term	Hadma-had	Почленно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $a > b$ and $c > d$ it follows that:

- $a + c > b + d$
- $a - c > b - d$
- $ac > bd$
- nothing

Multiplying two inequalities term by term needs:

- nothing extra
- all terms positive
- all terms whole numbers
- the same direction only

If $2 < a < 6$ then $-a$ satisfies:

- $-2 < -a < -6$
- $-6 < -a < -2$
- $2 < -a < 6$
- $-6 < -a < 6$

If $1 < b < 4$ then $\frac{1}{b}$ satisfies:

- $\frac{1}{4} < \frac{1}{b} < 1$
- $1 < \frac{1}{b} < 4$
- $\frac{1}{4} < \frac{1}{b} < \frac{1}{4}$
- it cannot be found

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $1 < a < 3$ and $2 < b < 5$. Find the range of $a + b$

   ANSWER $3 < a + b < 8$

2. $0 < a < 2$ and $1 < b < 4$. Find the range of $a + b$

   ANSWER $1 < a + b < 6$

3. $2 < a < 4$. Find the range of $-a$

   ANSWER $-4 < -a < -2$

4. $1 < a < 5$. Find the range of $2a$

   ANSWER $2 < 2a < 10$

5. $1 < a < 3$ and $2 < b < 4$. Find the range of ab

   ANSWER $2 < ab < 12$

6. $3 < a < 6$. Find the range of $a + 10$

   ANSWER $13 < a + 10 < 16$

7. *May two inequalities of the same direction be added?*

   ANSWER Yes.

Medium · the standard question

7 PROBLEMS

1. $3 < a < 5$ and $2 < b < 4$. Find the range of $a + b$

   ANSWER $5 < a + b < 9$

2. $3 < a < 5$ and $2 < b < 4$. Find the range of $a - b$

   ANSWER $-1 < a - b < 3$

3. $3 < a < 5$ and $2 < b < 4$. Find the range of ab

   ANSWER $6 < ab < 20$

4. $1 < b < 4$. Find the range of $\frac{1}{b}$

   ANSWER $\frac{1}{4} < \frac{1}{b} < 1$

5. $2 < x < 6$ and $1 < y < 3$. Find the range of $\frac{x}{y}$

   ANSWER $\frac{2}{3} < \frac{x}{y} < 6$

6. Show that $5 > 3$ and $10 > 1$ cannot be subtracted term by term

   ANSWER It would give $-5 > 2$, which is false.

7. $1 < a < 2$ and $3 < b < 5$. Find the range of $2a + b$

   ANSWER $5 < 2a + b < 9$

Hard · stretch and reason

7 PROBLEMS

1. $2 < a < 4$ and $1 < b < 3$. Find the range of $a^2 - b$

   ANSWER $4 < a^2 < 16$ and $-3 < -b < -1$, so $1 < a^2 - b < 15$

2. The sides of a rectangle satisfy $3 < a < 5$ and $4 < b < 6$. Estimate the perimeter.

   ANSWER $14 < P < 22$

3. The same rectangle: estimate the area.

   ANSWER $12 < S < 30$

4. Give a counter-example to “$a > b$ and $c > d$ implies $ac > bd$”.

   ANSWER $2 > -5$ and $3 > -4$, but $6 < 20$.

5. $1 < x < 3$. Find the range of $\frac{1}{x+1}$

   ANSWER $2 < x + 1 < 4$, so $\frac{1}{4} < \frac{1}{x+1} < \frac{1}{2}$

6. $2 < a < 3$ and $4 < b < 5$. Find the range of $\frac{b}{a}$

   ANSWER $\frac{1}{3} < \frac{1}{a} < \frac{1}{2}$, so $\frac{4}{3} < \frac{b}{a} < \frac{5}{2}$

7. $0 < a < 1$. Compare a^2 with a.

   ANSWER $a^2 < a$ — multiplying $a < 1$ by the positive number a.

07 Homework

Homework — 5 problems

Algebra 8, §13, pp. 75–79. Reverse the second range before any difference or quotient.

1. $4 < a < 7$ and $1 < b < 3$. Find the range of $a + b$
2. $4 < a < 7$ and $1 < b < 3$. Find the range of $a - b$
3. $4 < a < 7$ and $1 < b < 3$. Find the range of ab

4. *$4 < a < 7$ and $1 < b < 3$. Find the range of* $\frac{a}{b}$

5. *A rectangle has $2 < a < 4$ and $5 < b < 7$. Estimate its perimeter and its area.*